MTL106: Probability and Stochastic Processes
Lecture 12
January 28, 2026

Bivariate Continuous Random Variable and Joint PDF: A two-dimensional (or bivariate) random variable (X, Y) is said to be of the continuous type if there exists a nonnegative function $f(.,.)$ such that for every pair $(x, y) \in \mathbb{R}_2$, we have

$$F(x,y) = \int_{-\infty}^{x} \left[\int_{-\infty}^{y} f(u,v)dv \right] du, \tag{1}$$

where F is the CDF of (X, Y). The function f is called the *joint probability density function* (joint PDF) of (X, Y).

Clearly,

$$F(+\infty, +\infty) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(u,v)dvdu = 1.$$

If f is continuous at (x, y), then

$$\frac{\partial^2 F(x,y)}{\partial x \partial y} = f(x,y). \tag{2}$$

Example 1: Let (X, Y) be a random variable with joint PDF given by

$$f(x,y) = \begin{cases} e^{-x-y}, & 0 < x < \infty, \ 0 < y < \infty, \\ 0, & \text{otherwise.} \end{cases}$$

Then

$$\begin{aligned} F(x,y) &= \int_{0}^{x} \left[\int_{0}^{y} e^{-u-v} dv \right] du \\ &= \int_{0}^{x} [e^{-u} - e^{-u-y}] du \\ &= (1 - e^{-x})(1 - e^{-y}). \end{aligned}$$

$$\Rightarrow F(x,y) = \begin{cases} (1 - e^{-x})(1 - e^{-y}), & 0 < x < \infty, \ 0 < y < \infty, \\ 0, & \text{otherwise.} \end{cases}$$

Theorem 5. If f is a nonnegative function satisfying $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x,y)dxdy = 1$, then f is the joint density function of some random variable.

Marginal Probability Mass Function: Let (X, Y) be a two-dimensional random variable with PMF

$$p_{ij} = P(X = x_i, Y = y_j).$$

Then

$$\sum_{i=1}^{\infty} p_{ij} = \sum_{i=1}^{\infty} P\{X = x_i, Y = y_j\} = P(Y = y_j), \tag{3}$$

$$\sum_{j=1}^{\infty} p_{ij} = \sum_{j=1}^{\infty} P\{X = x_i, Y = y_j\} = P(X = x_j). \tag{4}$$

Let us write

$$p_{i.} = \sum_{j=1}^{\infty} p_{ij} \text{ and } p_{.j} = \sum_{i=1}^{\infty} p_{ij}. \tag{5}$$

Then $p_{i.} \geq 0$ and $\sum_{i=1}^{\infty} p_{i.} = 1$, $p_{.j} \geq 0$ and $\sum_{j=1}^{\infty} p_{.j} = 1$, and $\{p_{i.}\}, \{p_{.j}\}$ represent PMFs.

The collection of numbers $\{p_{i.}\}$ is called the marginal PMF of X, and the collection $\{p_{.j}\}$, the marginal PMF of Y.

Example 2: The joint PMF of (X, Y) is given by $P(X = x, Y = y) = \frac{2x+3y}{72}$, $x = 0, 1, 2$, $y = 1, 2, 3$. Find the marginal distributions for

1. $P(X = 2, Y \leq 2)$,
2. $P(X \leq 1, Y = 3)$,
3. $P(X = 2)$,
4. $P(X \leq 2)$.

Solution: Given that

		Y				
		1	2	3	**Total**	
	0	3/72	6/72	9/72	18/72	P(X=0)
X	1	5/72	8/72	11/72	24/72	P(X=1)
	2	7/72	10/72	13/72	30/72	P(X=2)
	Total	15/72	24/72	33/72	1	
	P(Y=1)	P(Y=2)	P(Y=3)			

1. $P(X = 2, Y \leq 2) = P(X = 2, Y = 1) + P(X = 2, Y = 2) = 7/72 + 10/72 = 17/72.$

2. $P(X \le 1, Y=3) = P(X=0, Y=3) + P(X=1, Y=3) = 9/72 + 11/72 = 20/72.$

3. $P(X=2) = \sum_{j=1}^{3} p_{2j} = P(X=2, Y=1) + P(X=2, Y=2) + P(X=2, Y=3)$
   =7/72+10/72+13/72=30/72.

4. $P(X \le 2) = \sum_{j=1}^{3} p_{0j} + \sum_{j=1}^{3} p_{1j} + \sum_{j=1}^{3} p_{2j}$
   $= P(X=0) + P(X=1) + P(X=2) = 18/72 + 24/72 + 30/72 = 72/72 = 1.$

Marginal Probability Density Function: If (X, Y) is a random variable of the continuous type with PDF f, then

$$f_1(x) = \int_{-\infty}^{\infty} f(x,y)dy \tag{6}$$

and

$$f_2(y) = \int_{-\infty}^{\infty} f(x,y)dx \tag{7}$$

satisfy $f_1(x) \ge 0,\ \ f_2(y) \ge 0$, and $\int_{-\infty}^{\infty} f_1(x)dx = 1,\ \ \int_{-\infty}^{\infty} f_2(y)dy = 1$. It follows that $f_1(x)$ and $f_2(y)$ are PDFs.

The functions $f_1(x)$ and $f_2(y)$, defined in (6) and (7), are called the marginal probability density functions of X and Y, respectively.

Example 3. Let (X, Y) be jointly distributed with PDF

$$f(x,y) = \begin{cases} 2, & 0 < x < y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Then

$$f_1(x) = \int_{x}^{1} 2dy = \begin{cases} 2 - 2x, & 0 < x < 1, \\ 0, & \text{otherwise,} \end{cases}$$

and

$$f_2(y) = \int_{0}^{y} 2dx = \begin{cases} 2y, & 0 < y < 1, \\ 0, & \text{otherwise} \end{cases}$$

are two marginal density functions.

Marginal Cumulative Distribution Function: Let (X, Y) be a random variable with CDF

F. Then the *marginal cumulative distribution function* of X is given by

$$F_1(x) = F(x, \infty) = \lim_{y \to \infty} F(x, y) = \begin{cases} \sum\limits_{x_i \leq x} p_i, & \text{if } (X, Y) \text{ is discrete,} \\ \int\limits_{-\infty}^{x} f_1(t)dt, & \text{if } (X, Y) \text{ is continuous.} \end{cases} \tag{8}$$

Similarly one can obtain the marginal PDF of Y.